

PROBLEMS

Econ 502: Advanced Microeconomics

Problem 1: Monopoly Pricing

Part (a): Monopoly equilibrium

Total revenue is $TR = P \cdot Q = (200 - 2Q)Q = 200Q - 2Q^2$. Marginal revenue:

$$MR = \frac{dTR}{dQ} = 200 - 4Q$$

Setting $MR = MC$:

$$200 - 4Q = 40 \implies Q_m^* = 40$$

$$P_m^* = 200 - 2(40) = 120$$

$$\pi = (P - MC) \cdot Q = (120 - 40)(40) = \$3,200$$

$$\boxed{Q_m^* = 40, \quad P_m^* = \$120, \quad \pi = \$3,200}$$

Part (b): Competitive equilibrium

Under perfect competition, $P = MC$:

$$200 - 2Q = 40 \implies Q_c = 80, \quad P_c = \$40$$

Consumer surplus is the area of the triangle between the demand curve and the price:

$$CS_c = \frac{1}{2}(200 - 40)(80) = \$6,400$$

Producer surplus is zero since the firm earns no markup ($P = MC$ with constant marginal cost). Total surplus:

$$TS_m = (120 - 40)(40) = \$3,200$$

$$TS_m = 1,600 + 3,200 = \$4,800$$

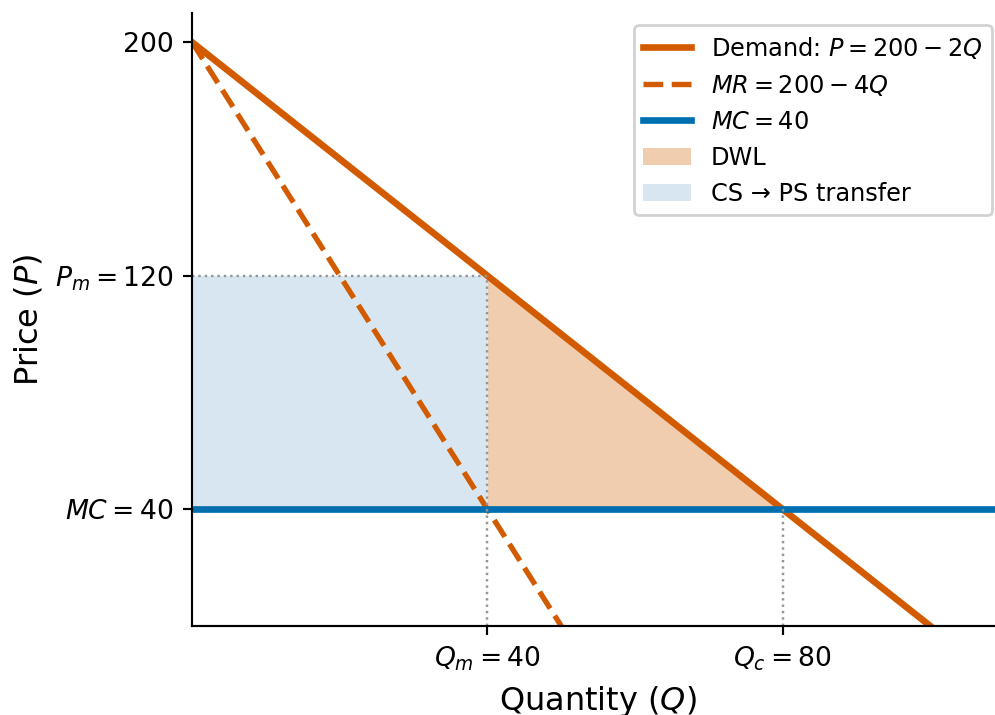
The deadweight loss is the surplus destroyed by monopoly pricing:

$$DWL = TS_c - TS_m = 6,400 - 4,800 = \$1,600$$

Equivalently, the DWL is the triangle between the demand curve and MC over the range of output lost:

$$DWL = \frac{1}{2}(P_m - MC)(Q_c - Q_m) = \frac{1}{2}(80)(40) = \$1,600$$

$$DWL = \$1,600$$



Part (d): Lerner Index and the inverse elasticity rule

The Lerner Index measures the firm's markup as a fraction of price:

$$\varepsilon_D = \frac{dQ}{dP} \cdot \frac{P}{Q} = -\frac{1}{2} \cdot \frac{3}{40} = -\frac{3}{80}$$

The inverse elasticity rule states that $L = 1/|\varepsilon_D|$:

$$\frac{1}{|\varepsilon_D|} = \frac{1}{3/80} = \frac{80}{3} \quad \checkmark$$

$$L = \frac{80}{3}, \quad |\varepsilon_D| = \frac{3}{80}$$

The monopolist operates in the elastic portion of the demand curve ($|\varepsilon_D| > 1$), as we would expect. If demand were inelastic, the firm could increase revenue by raising the price (reducing output), so it would never be optimal to produce in the inelastic region.

Problem 2: Third-Degree Price Discrimination

Part (a): Price discrimination

The inverse demand curves are $P_A = 100 - Q_A$ and $P_S = 60 - Q_S$. The monopolist maximizes profit in each market separately by setting $MR = MC$:

Adults:

$$MR_A = 100 - 2Q_A = 20 \implies Q_A = 40, \quad P_A = 60$$

$$\pi_A = (60 - 20)(40) = \$1,600$$

Students:

$$MR_S = 60 - 2Q_S = 20 \implies Q_S = 20, \quad P_S = 40$$

$$\pi_S = (40 - 20)(20) = \$400$$

$$\text{Total profit} = 1,600 + 400 = \$2,000, \quad \text{Total output} = 60$$

Check that both markets are served: $Q_A = 100 - 50 = 50 > 0$ and $Q_S = 60 - 50 = 10 > 0$. ✓

$$P = \$50, \quad Q = 60, \quad \pi = (50 - 20)(60) = \$1,800$$

Note that total output is the same under both regimes (60 units). This is a general result for linear demand curves when both markets are served under uniform pricing.

Part (c): Elasticities and Lerner index

Under price discrimination, at the optimal prices:

$$\text{Adults: } \varepsilon_A = \frac{dQ_A}{dP_A} \cdot \frac{P_A}{Q_A} = (-1) \cdot \frac{60}{40} = -\frac{3}{2}$$

$$L_A = \frac{P_A - c}{P_A} = \frac{60 - 20}{60} = \frac{2}{3} = \frac{1}{|\varepsilon_A|} \quad \checkmark$$

$$\text{Students: } \varepsilon_S = \frac{dQ_S}{dP_S} \cdot \frac{P_S}{Q_S} = (-1) \cdot \frac{40}{20} = -2$$

$$L_S = \frac{P_S - c}{P_S} = \frac{40 - 20}{40} = \frac{1}{2} = \frac{1}{|\varepsilon_S|} \quad \checkmark$$

$$L_A = \frac{2}{3} > L_S = \frac{1}{2}$$

Adults get the higher markup because their demand is less elastic ($|\varepsilon_A| = 3/2 < |\varepsilon_S| = 2$). Adults are less price-sensitive, so the monopolist can extract more surplus from them by charging a higher price relative to marginal cost. This is the key insight of third-degree price discrimination: charge more where demand is less elastic.

Part (d): Welfare comparison

Under price discrimination:

Total CS = 1,300, Profit = 1,800, Total welfare = 3,100

Welfare is \$100 higher under uniform pricing

	CS	Profit	Total Welfare
Price discrimination	1,000	2,000	3,000
Uniform pricing	1,300	1,800	3,100

Price discrimination is **not** welfare-improving here. Consumers are worse off (-300 in CS) and the firm is better off ($+200$ in profit), but the net effect is a welfare loss of 100. Since total output is the same under both regimes, discrimination simply reallocates output from the more elastic market (students) to the less elastic market (adults). This reallocation is inefficient: units are moved from consumers with higher marginal valuations (students priced out at $P_S = 40$) to consumers with lower marginal valuations (additional adults served at $P_A = 60$).

Note: If the student market would be shut out entirely under uniform pricing (e.g., if c were higher), price discrimination could be welfare-improving by opening a new market.

Problem 3: Cournot Duopoly with Asymmetric Costs

Part (a): Best response functions

Firm 1 ($MC_1 = 20$) maximizes profit given q_2 :

$$\pi_1 = (120 - q_1 - q_2)q_1 - 20q_1 = (100 - q_1 - q_2)q_1$$

$$\frac{\partial \pi_1}{\partial q_1} = 100 - 2q_1 - q_2 = 0 \implies q_1^*(q_2) = 50 - \frac{q_2}{2}$$

Firm 2 ($MC_2 = 40$) maximizes profit given q_1 :

Part (b): Nash equilibrium

Solve the two best response equations simultaneously. Substitute

$q_2^* = 40 - q_1/2$ into q_1^* :

$$q_1 = 50 - \frac{1}{2} \left(40 - \frac{q_1}{2} \right) = 50 - 20 + \frac{q_1}{4} = 30 + \frac{q_1}{4}$$

$$\frac{3q_1}{4} = 30 \implies q_1^* = 40$$

$$q_2^* = 40 - \frac{40}{2} = 20$$

$$Q^* = 60, \quad P^* = 120 - 60 = 60$$

$$\pi_1^* = (60 - 20)(40) = \$1,600, \quad \pi_2^* = (60 - 40)(20) = \$400$$

$$\boxed{q_1^* = 40, q_2^* = 20, P^* = \$60, \pi_1^* = \$1,600, \pi_2^* = \$400}$$

Part (c): Lerner index and market power

$$L_1 = \frac{P^* - MC_1}{P^*} = \frac{60 - 20}{60} = \frac{2}{3}, \quad L_2 = \frac{P^* - MC_2}{P^*} = \frac{60 - 40}{60} = \frac{1}{3}$$

Firm 1 has more market power ($L_1 = 2/3 > L_2 = 1/3$).

To understand why, note that in Cournot equilibrium each firm's Lerner index equals its market share divided by the market price elasticity of demand:

$$L_i = \frac{s_i}{|\varepsilon_D|}, \quad \text{where } s_i = \frac{q_i^*}{Q^*}, \quad |\varepsilon_D| = \frac{P^*}{Q^*} = \frac{60}{60} = 1$$

$$s_1 = \frac{40}{60} = \frac{2}{3}, \quad s_2 = \frac{20}{60} = \frac{1}{3} \implies L_1 = \frac{2}{3}, \quad L_2 = \frac{1}{3} \checkmark$$

Firm 1's lower marginal cost lets it produce more output and capture a larger market share, which in turn gives it greater price-setting power. The Lerner index reflects market share, not cost advantage directly.

This is a general result for linear Cournot models: with n firms and inverse demand $P = a - bQ$,

$$Q^* = \frac{na - \sum_i MC_i}{(n+1)b}$$

Both cases have $\sum MC_i = 20 + 40 = 30 + 30 = 60$, so $Q^* = (2 \times 120 - 60)/3 = 60$ and $P^* = 60$ in both.

Since P and Q are the same, **consumer surplus is identical**. The difference lies in production costs:

	q_1	q_2	Total cost	Total profit	Total welfare
Asymmetric ($MC_1 = 20, MC_2 = 40$)	40	20	$20(40) + 40(20) = 1,600$	2,000	3,800
Symmetric ($MC_1 = MC_2 = 30$)	30	30	$30(30) + 30(30) = 1,800$	1,800	3,600

(Total welfare = CS + total profit = 1,800 + total profit.)

Asymmetric costs generate **higher total welfare** by \$200. The Cournot mechanism reallocates production from the high-cost firm (Firm 2 drops from 30 to 20 units) to the low-cost firm (Firm 1 rises from 30 to 40 units). This productive reallocation reduces total production cost for the same output, raising total surplus. In other words, when cost differences exist, even imperfect competition partially corrects the production inefficiency of a symmetric allocation.

Problem 4: Bertrand Competition

Part (a): $p_1 = p_2 = c$ is a Nash equilibrium

We need to show that neither firm can increase its profit by unilaterally changing its price.

Since no deviation is profitable, $p_1 = p_2 = c$ is a Nash equilibrium. ✓

Part (b): $p_1 = p_2 > c$ is not a Nash equilibrium

Suppose both firms charge $p = p_1 = p_2 > c$, splitting the market equally. Each firm earns:

$$\pi = (p - c) \cdot \frac{Q(p)}{2} > 0$$

Now consider firm 1 deviating to $p_1 = p - \epsilon$ (for small $\epsilon > 0$). Since the good is homogeneous and firm 1 is now cheaper, it captures the **entire market**:

$$\pi_1^{\text{dev}} = (p - \epsilon - c) \cdot Q(p - \epsilon)$$

For small ϵ , this is approximately $(p - c) \cdot Q(p)$, which is roughly **double** firm 1's original profit. Since this is a profitable deviation, the original prices cannot be a Nash equilibrium. ✓

Part (c): Asymmetric costs

With $MC_1 = 10$ and $MC_2 = 20$, firm 1 has a cost advantage.

If firm 1 sets $p_1 = 20 - \epsilon$ (just below firm 2's cost), it captures the entire market while firm 2 cannot profitably undercut (doing so would mean pricing below its own marginal cost).

At $p_1 = 20$ (taking the limit as $\epsilon \rightarrow 0$):

$$Q = 100 - 20 = 80$$

$$\pi_1 = (20 - 10)(80) = \$800$$

Firm 2 earns \$0.

$p_1^* = 20, \quad \pi_1 = \$800, \quad \pi_2 = \$0$
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Problem 5: Differentiated Bertrand Competition

Part (a): Best response functions

Firm 1 maximizes $\pi_1 = (p_1 - 1)(10 - 2p_1 + p_2)$. The FOC with respect to p_1 is:

$$\frac{\partial \pi_1}{\partial p_1} = (10 - 2p_1 + p_2) + (p_1 - 1)(-2) = 0$$

$$10 - 4p_1 + p_2 + 2 = 0 \implies p_1^*(p_2) = 3 + \frac{p_2}{4}$$

By symmetry, $p_2^*(p_1) = 3 + p_1/4$.

Prices are **strategic complements**: the best response is upward sloping ($\partial p_1^* / \partial p_2 = 1/4 > 0$). If firm 2 raises its price, firm 1's demand rises, and it optimally charges more as well.

Part (b): Nash equilibrium

In a symmetric equilibrium $p_1 = p_2 = p^*$:

$$p^* = 3 + \frac{p^*}{4} \implies \frac{3p^*}{4} = 3 \implies p^* = 4$$

$$q_i = 10 - 2(4) + 4 = 6, \quad \pi_i = (4 - 1)(6) = \$18$$

$$p^* = 4, \quad q_i = 6, \quad \pi_i = \$18$$

Part (c): Why equilibrium price exceeds marginal cost

$p^* = 4 > c = 1$, so firms earn positive profits in equilibrium. With **product differentiation**, a firm that slightly undercuts its rival does not capture the

$$\Pi = 2(p - 1)(10 - 2p + p) = 2(p - 1)(10 - p)$$

FOC:

$$\frac{d\Pi}{dp} = 2[(10 - p) - (p - 1)] = 2(11 - 2p) = 0 \implies p^{\text{col}} = 5.5$$

$$q_i^{\text{col}} = 10 - 2(5.5) + 5.5 = 4.5, \quad \pi_i^{\text{col}} = (5.5 - 1)(4.5) = \$20.25$$

$$p^{\text{col}} = 5.5 > p^* = 4, \quad \pi_i^{\text{col}} = \$20.25 > \pi_i^* = \$18$$

Collusion raises prices and profits for both firms. Each firm has an incentive to agree on the higher price but also to cheat by undercutting, which is the classic tension that makes collusion unstable without a credible enforcement mechanism.

Problem 6: Stackelberg Competition

Part (a): Cournot equilibrium

Firm 1 maximizes $\pi_1 = (150 - q_1 - q_2 - 30)q_1 = (120 - q_1 - q_2)q_1$.

FOC: $120 - 2q_1 - q_2 = 0 \implies q_1^*(q_2) = 60 - q_2/2$. By symmetry, $q_2^*(q_1) = 60 - q_1/2$.

Solving simultaneously:

$$q_i^C = 40, \quad Q^C = 80, \quad P^C = 70, \quad \pi_i^C = (70 - 30)(40) = \$1,600$$

$$q_i^C = 40, \quad P^C = \$70, \quad \pi_i^C = \$1,600$$

Part (b): Stackelberg equilibrium

Step 1: Follower's best response (same as Cournot):

$$q_2^*(q_1) = 60 - \frac{q_1}{2}$$

$$\pi_1^S = (60 - 30)(60) = \$1,800, \quad \pi_2^S = (60 - 30)(30) = \$900$$

$$q_1^S = 60, q_2^S = 30, P^S = \$60, \pi_1^S = \$1,800, \pi_2^S = \$900$$

Part (c): Comparison and the top dog strategy

	Cournot	Stackelberg Leader	Stackelberg Follower
Output	40	60	30
Price	\$70	\$60	\$60
Profit	\$1,600	\$1,800	\$900

The leader gains from moving first (profit rises from \$1,600 to \$1,800), while the follower is worse off (\$900 < \$1,600). By committing to a large quantity before the follower acts, the leader exploits the fact that quantities are **strategic substitutes** – a higher q_1 shifts the follower's best response downward, inducing firm 2 to produce less. This credible commitment to “overproducing” is the “**top dog**” strategy: be aggressive to force the rival to back off.

Note that firm 1's Stackelberg output (60) is exactly its monopoly output; the leader takes the entire “first half” of the market, leaving the follower to split the remainder.

Part (d): Consumer surplus and welfare

$$CS^C = \frac{1}{2}(150 - 70)(80) = \frac{1}{2}(80)(80) = \$3,200$$

$$TS^C = CS^C + \pi_1^C + \pi_2^C = 3,200 + 1,600 + 1,600 = \$6,400$$

$$CS^S = \frac{1}{2}(150 - 60)(90) = \frac{1}{2}(90)(90) = \$4,050$$

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price down from \$70 to \$60, raising CS by \$850. Total welfare is also higher (\$6,750 vs \$6,400), despite the fall in industry profit (\$2,700 vs \$3,200) — the consumer surplus gain more than offsets the profit loss. Stackelberg moves the market outcome closer to the competitive benchmark.